

## Experiment 21

### Title

### Write a Prolog Program to implement Towers of Hanoi

### Aim

To implement the Towers of Hanoi problem using Prolog and display the sequence of moves required to transfer all the disks from the source peg to the destination peg.

### Objective

To understand the concept of recursion in Prolog by solving the Towers of Hanoi problem and generating the correct sequence of disk movements.

### Algorithm

**Step 1:** Define a base case to move one disk from the source peg to the destination peg.

**Step 2:** If the number of disks is greater than one, move  $N-1$  disks from the source peg to the auxiliary peg.

**Step 3:** Move the largest disk from the source peg to the destination peg.

**Step 4:** Move the  $N-1$  disks from the auxiliary peg to the destination peg.

**Step 5:** Display each move until all the disks are transferred successfully.

### Program

```
% Towers of Hanoi
```

```
hanoi(1, Source, _, Destination) :-
```

```
    write('Move disk 1 from '),
```

```
    write(Source),
```

```
    write(' to '),
```

```
    write(Destination), nl.
```

```
hanoi(N, Source, Auxiliary, Destination) :-
```

```
    N > 1,
```

```
    M is N - 1,
```

```
    hanoi(M, Source, Destination, Auxiliary),
```

```
    write('Move disk '),
```

```
write(N),  
write(' from '),  
write(Source),  
write(' to '),  
write(Destination), nl,  
hanoi(M, Auxiliary, Source, Destination).
```

### Sample Input

?- hanoi(3, 'A', 'B', 'C').

### Sample Output

```
| ?- consult('C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp21.pl').  
compiling C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp21.pl for byte code...  
C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp21.pl compiled, 18 lines read - 1718 bytes written, 4 ms  
  
(31 ms) yes  
| ?- hanoi(3, 'A', 'B', 'C').  
Move disk 1 from A to C  
Move disk 2 from A to B  
Move disk 1 from C to B  
Move disk 3 from A to C  
Move disk 1 from B to A  
Move disk 2 from B to C  
Move disk 1 from A to C  
  
true ?  
  
(31 ms) yes
```

### Result

Hence, the Prolog program to implement the **Towers of Hanoi** was executed successfully and the correct sequence of moves was obtained.